A Review of Signal Processing Techniques for Electrocardiogram Signal Quality Assessment

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***Abstract*—Electrocardiogram (ECG) signal quality assessment (SQA) plays a vital role in significantly improving the diagnostic accuracy and reliability of unsupervised ECG analysis systems. In practice, the ECG signal is often corrupted with different kinds of noises and artifacts. Therefore, numerous SQA methods were presented based on the ECG signal and/or noise features and the machine learning classifiers and/or heuristic decision rules. This paper presents an overview of current state-of-the-art SQA methods and highlights the practical limitations of the existing SQA methods. Based upon past and our studies, it is noticed that a lightweight ECG noise analysis framework is highly demanded for real-time detection, localization, and classification of single and combined ECG noises within the context of wearable ECG monitoring devices which are often resource constrained.**

***Index Terms*—Electrocardiogram (ECG), ECG noise, signal quality assessment, signal quality index.**

I. Introduction

ELECTROCARDIOGRAM (ECG) signals are widely acquired for many applications such as cardiovascular disease diagnosis, arrhythmias recognition, physiological feedback, sleep apnea detection, chronic patient surveillance, sudden cardiac arrest prediction, biometric, emotional, and physical activity recognition systems [1]–[6]. All of these ECG application systems highly demand the exact determination of fiducial points (or characteristic points) of the ECG signal for accurate and reliable measurements of morphological features (including, amplitude, duration, polarity, and shape) of local waves such as P-wave, QRS-complex, T-wave and U-wave, and the interval features (including, PR-segment, QT-segment, ST-segment, RR interval) [4]–[6]. Most ECG analysis systems require relatively noise-free ECG signals for obtaining the ECG measurements more accurately and reliably [6]. In practice, the ECG signals are often corrupted with different types of noises such as baseline wander (electrode contact noise and electrode motion artifacts), power line interference (PLI), electromyogram (EMG) noise, and instrumentation noise, making it almost impossible to perform a morphological analysis of such contaminated ECG beats [6]–[11]. Therefore, automatic assessment of ECG signal is highly demanded in reducing the false alarms due to presence of unacceptable level of noises.

Manuscript received June 25, 2017; revised December 7, 2017 and February 25, 2018; accepted February 25, 2018. Date of publication February 28, 2018; date of current version July 24, 2018. *(Corresponding author: Udit Satija.)*

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Digital Object Identifier 10.1109/RBME.2018.2810957

A. Characteristics of ECG Noises

A brief description of temporal and spectral characteristics of ECG noise sources is summarized below.

1) Baseline Wander and Abrupt Drift: The baseline wander and abrupt drift noises are caused due to the respiration, body movements, poor electrode contact, and skin-electrode impedance [8], [9]. The peak amplitude and duration of the baseline wander (BW) depend on the various factors such as the electrode properties, electrolyte properties, skin impedance, and the subject movement [8]. The frequency spectrum of the baseline wandering ranges between 0.05 and 1 Hz. The amplitude of the respiration baseline wander varies by about 15% of peak-to-peak ECG amplitude at frequencies between 0.15 and 0.3 Hz [8]. Under stress ECG recordings, the frequency of the baseline wandering increases due to the increased breathing rate. The motion artifacts are transient baseline changes (rapid drifts) caused by changes in the electrode-skin impedance with electrode motion [9]. The magnitude of the baseline wander may exceed the amplitude of the QRS complex by several times. The amplitude and duration of the motion artifact are 500% of peak–peak ECG amplitude and 300–500 ms, respectively [9]. The severe baseline wanders or motion artifacts can distort the ST-segment and other low-frequency components of ECG signal. The ST-segment distortion may lead to wrong diagnosis of myocardial infarction, Brugada syndrome, and other ST-segment related abnormalities.

2) Power-Line Interference: The PLI noises are caused by inductive and capacitive couplings of ubiquitous power lines in the ECG signal acquisition circuitry. The PLI is a narrow-band noise centered at 50/60 Hz with a bandwidth of less than 1 Hz and an amplitude of up to 50% of the peak-to-peak ECG amplitude [10]. The lower frequency noise components of the PLI are mixed with the frequency content of the ECG signal. The severe structured noises can distort the morphological features such as amplitude, duration, and shape of low-amplitude local waves of the ECG signal. In particularly, the P-wave distortions can lead to wrong diagnosis of atrial arrhythmias such as atrial enlargement and fibrillation.

3) Muscle Artifacts: The muscle artifacts (or EMG) are caused by the electrical activity of muscles during periods of

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contraction or due to a sudden body movement. In general, the amplitude of an EMG noise is about 10% of the ECG amplitude with a bandwidth between 20 and 1000 Hz. Some studies show that the frequency range is between dc and 10 000 Hz with an average amplitude of 10% level [8]. Previous studies showed that the muscle artifacts can significantly alter the shapes of local waves of the ECG signal because the frequency of EMG noise considerably overlaps with that of the ECG signals, in the range of 0.01–100 Hz [9]. Thus, removal of muscle artifacts is quite challenging without distorting the clinical features, which are most essential for recognizing various ECG arrhthmias [12].

B. Issues of Automated ECG Analysis Under Noisy Conditions

Literature studies showed that existing ECG analysis systems render the inaccurate and unreliable measurements, which lead to produce high false alarm rates for the noisy ECG signals [11]–[14]. Frequent false alarms are not only most annoying and disturbing both the clinicians and the patients but also lead to misdiagnosis of cardiac arrhythmias [13], [14]. The existing ECG event detection and recognition algorithms still suffer from false alarm rates because of the failure to distinguish between the ECG noises and artifacts and the real ECG morphology. Therefore, noisy ECG signals must either be discarded or filtered prior to the feature extraction using noise-specific signal processing schemes.

The issue of high false arrhythmia event alarm and heart-rate alarm rates highly impacts the usability of real-time ECG monitoring systems because of the two main reasons: first, Noises and artifacts in the isoelectric line of the ECG signal are falsely detected as normal beats or abnormal beats; and Second, severely contaminated ECG beats are misclassified due to the inaccurate measurements of essential ECG feature parameters. Therefore, various strategies have been adopted to handle the problem of high false alarm rate due to the noises and artifacts: 1) *ECG denoising based strategy* to suppress the noises and artifacts in the ECG recordings and 2) *Signal quality index (SQI) based strategy* to assess the clinical acceptability of the recorded ECG signals [11]–[14].

1) ECG Denoising Based Strategy: To reduce the false alarm rates, various ECG denoising methods have been presented based on the moving average and median filters, frequency-selective filters, adaptive filters, Wiener filters, polynomial filters, singular value decomposition (SVD), discrete cosine transform (DCT), discrete wavelet transform (DWT), empirical mode decomposition (EMD), nonlinear Bayesian filter, mathematical morphological operators, independent component analysis (ICA), nonlocal means method, variational mode decomposition and EMD-wavelet method for removal of single and combined ECG noise sources [11], [12], [15], [16]. In general, the denoising methods are highly capable of suppressing noises and artifacts that are very different from those of ECG morphology. Evaluation results show that the baseline removal methods may distort the ST-segment due to the attenuation of the low-frequency components of the ECG signal [16]. Results further show that simple filters are not adequate to remove severe EMG noise without distorting the amplitude, duration,

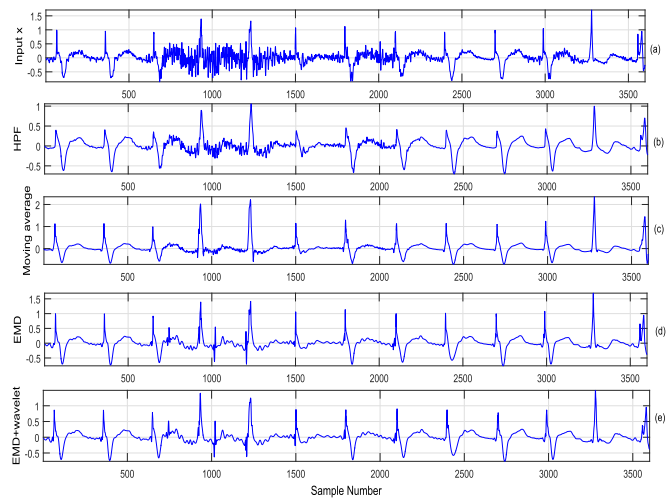

Fig. 1. Illustrates the noise reduction performance of the denoising methods such as high-pass filter, moving average filter, EMD and EMD plus DWT [12].

interval, and shape features [12]. Results show that the EMD-based denoising method introduces significant distortion at the beginning and end of QRS complexes that may cause widening of the QRS complex. The adaptive and nonadaptive filters introduce significant distortion in the QRS and ST-segment portions due to the filter ringing problem [12]. The noise reduction capability of four denoising methods is shown in Fig. 1 for the ECG signal with localized muscle artifacts as reported in [12]. The existing denoising methods showed that severe motion artifacts cannot be removed effectively using linear filters. However, some studies show that most denoising methods significantly alter the ECG local waves due to the impossibility of handling the problem of ECG noises and artifacts that considerably overlaps with spectrum of ECG signals, in the range of 0.5–100 Hz [12]. Previously published works have emphasized on noise reduction, but none of them have proposed an approach to handle the problem of muscle artifacts that mimic the ECG waveform morphology. Additionally, existing methods may distort the ECG morphology after noise suppression. Moreover, a single signal processing technique may not be adequate to remove different kinds of single and combined ECG noises. Furthermore, each of the denoising techniques distorts different local waves such as P-wave, QRS-complex, and T-wave, and also introduces different kinds of waveform distortions such as amplitude reduction, width widening and shape alteration. In such scenarios, the normal heartbeats are misclassified as abnormal heartbeats (or life-threatening heartbeats) and vice-versa due to the R-peak detection errors (missed heartbeats, erroneously detected heartbeats) and ECG waveform delineation errors. Thus, some of the heartbeat classification methods utilized the ECG signal quality assessment (SQA) strategy for distinguishing between good- and bad-quality ECG signals.

2) SQI Based Strategy: Besides the noise reduction strategy, SQA (or SQI) strategy has been employed to tackle the false alarm problem. Several signal quality assessment methods have been introduced for assessing the quality of ECG signals before feature extraction [17]–[21]. Most methods were presented for

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classifying the recorded ECG signals into acceptable or unacceptable [17]–[21]. In addition to the physiological features, an extensive set of SQI is computed from the raw ECG signal. These SQIs are then fused heuristically (or used as weighting factors) to reduce the false alarm rates in most ECG diagnostic systems.

This paper presents an overview of different signal processing techniques and classifiers used for assessing the quality of ECG signals. We further study the performance of some of the SQA methods using the standard ECG databases including different kinds of noises and pathological heartbeat patterns. The rest of this paper is organized as follows. Section II presents various test ECG databases and performance evaluation metrics. Section III describes in detail the different ECG SQA methods. Section IV presents performance evaluation results of some of the SQA methods. Section V highlights challenging issues and future directions. Finally, conclusions are drawn in Section VI.

II. Test ECG Databases and Performance Metrics

A. Test ECG Databases

Most ECG analysis methods are evaluated using the publicly available ECG databases including, the MIT-BIH Arrhythmia (MITBIHA) database [22], the PhysioNet/Computing in Cardiology Challenge 2011 Database [23], the MIT-BIH ST Change (MIT-BIHSTC) database [24], the Fantasia database [25], the Motion Artifact Contaminated ECG (MACE) database [26], the MIT-BIH Noise stress test (MITBIHNST) database [27], the MIMIC II database [28], and the TELE ECG database [29], [30]. For each of the ECG databases, the specifications of ECG recordings are summarized in Table I. The ECG databases contain different types of PQRST morphological patterns and various kinds of artifacts and noises. Some of the methods recorded the ECG signals under different recording conditions using the teleMedCare health monitor (TMC-HM).

B. Existing ECG Signal Quality Groups

In general, the SQA is described as "the process of determining clinical acceptability of the ECG signals, which can be freed from the various noises and artifacts introduced during different condition of ECG recordings." In the previously published works, the ECG signal quality is categorized into following quality groups.

1) Five Quality Group: *Excellent, Good, Adequate, Poor, and Unacceptable* [43].
2) Three Quality Group: *Acceptable, Indeterminate, and Unacceptable* [44]–[46], [61], [67]
3) Two Quality Group: *Acceptable and Unacceptable* [31]–[41], [47]–[60].

In the literature, the two quality group approach is most widely adopted in determining the ECG quality for further processing.

C. Performance Evaluation Metrics

The performance of the SQA methods is evaluated using the five benchmark metrics such as sensitivity (Se), positive predictivity (Pp), negative predictivity (Np), specificity (Sp), and accuracy (Ac) [40]–[48], which are defined as

$$\mathrm{Se} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}}, \qquad \mathrm{Sp} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FP}} \tag{1}$$

$$\mathrm{Pp} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}}, \qquad \mathrm{Np} = \frac{\mathrm{TN}}{\mathrm{TN}+\mathrm{FN}} \tag{2}$$

$$\mathrm{Ac} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{3}$$

where the true positive (TP) denotes a number of correctly identified unacceptable or noisy ECG signals by a SQA method, the false negative (FN) denotes a number of inadequate quality of ECG signals, which are identified as the adequate signal quality by the method, the true negative (TN) denotes a number of correctly identified acceptable ECG signals, and the false positive (FP) denotes a number of acceptable ECG quality signals which are identified as unacceptable quality by the SQA method.

III. ECG SQA Methods

In this paper, the SQA methods are grouped into five categories:

1) Fiducial features and heuristic rules based SQA methods;
2) Fiducial features and machine learning based SQA methods;
3) Nonfiducial features and heuristic rules based SQA methods;
4) Nonfiducial features and machine learning based SQA methods; and
5) Filtering-based SQA methods.

Most methods generally consist of three major stages:

1) *preprocessing* stage, which generally performs signal/noise filtering task;
2) *feature extraction* stage, which extracts the ECG signal and/or noise features; and
3) *classification* stage, which classifies the signal into two or more ECG signal quality groups.

The *preprocessing stage* employs various types of digital filters and decomposition techniques for discriminating different kinds of noise components from ECG signal components. The *feature extraction stage* is designed to extract the nonfiducial features (including, time-domain, frequency-domain, time-frequency domain, statistical, and information theoretic features) and fiducial features (including, ECG waveform morphological and heartbeat interval features). The *classification stage* uses machine learning techniques and decision rules (or heuristic rules) for grading the ECG signal. Each of the SQA methods is briefly described in the next sections by considering the key developmental resource-energy constraints of wearable ECG monitoring systems under unavoidable noisy recording environments.

A. Fiducial Features and Heuristic Rules Based Methods

In fiducial features and heuristic rules (FF-HR) based SQA methods, a variety of morphological and interval features (including, the duration and amplitude of P-wave, QRS-complex

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TABLE I
DESCRIPTION OF TEST ECG DATABASES

ECG Databases	No. of Records	Duration	Sampling Frequency	Bits/sample	No. of Leads	Heartbeat Types	Noise Types	Remarks
MITBIHA [22]	48	30 min	360 Hz	11	2	NSR, PVC, APC, LBBB, RBBB, P, VF, F, j, f, x, and Q	BW, MA, PLI, EM, AB	Mostly records are noise-free
PICC 2011 [23]	2000	10 s	500 Hz	16	12	–	FL, Spk, AB, BW, MA, PLI, EM	Contains both acceptable and unacceptable category of data
MIT-BIHSTC [24]	28	24 min	360 Hz	12	2	Myocardial hypertrophy, Valvular heart disease, Myocarditis, Miscellaneous, Healthy controls	MA, PLI, BW	Mostly noise-free
Fantasia [25]	40	120 min	250 Hz	16/12	1	NSR	MA, PLI, BW	Mostly noise-free
MACE [26]	27	8 s	500 Hz	16	1	NSR	Motion artifact	Contains motion artifact introduced in different activities
MIHBIHNST [27]	2	30 min	360	11	2	NSR, RBBB, APC, PVC, Blocked APC	MA, EM, and BW	Noise is added at different SNRs
MIMIC-II [28]	>800	–	125	10	–	Multiple expert-annotations for alarms for asystole, extreme bradycardia, extreme tachycardia, and ventricular tachycardia	–	–
TELE [29], [30]	250	48 min	500	12	1	Pathological segments	MA, EM	Contains large set of bad quality signals

Normal sinus rhythm, NSR; premature ventricular contraction, PVC; atrial premature contraction, APC; left bundle branch block, LBBB; right bundle branch block, RBBB; paced beat, P; ventricular fibrillation, VF; fusion of ventricular and normal beat, F; nodal escape (junctional) beat, j; fusion of paced and normal beat, f; nonconducted P-wave, x; unclassified beat, Q; baseline wander, BW; muscle artifact, MA; electrode motion, EM; PLI, PLI; Flatline, FL; Spike, Spk;

and T-wave, PR and ST-segment and intervals, QT and RR intervals) were extracted from single or multi-lead ECG signals and then combined with heuristic rules with predefined decision thresholds. Table II summarizes some of fiducial features and heuristic rules based SQA methods.

1) SOI for ECG and PPG Signals: Orphanidou *et al.* [31] proposed signal quality indices (SQIs) for determining the reliable heart rates. The signal quality was assessed using the heartbeat features and template matching. The method was evaluated on the PICC database and ambulatory ECG records available from John Radcliffe (JR) hospital in Oxford. The method had a Sp of 97% and a Se of 93%. It is noted that the performance of the SQA method highly relies on accurate and reliable QRS complex detection under ambulatory and exercise ECG recording conditions.

2) Real-Time Ambulatory ECG Quality Analysis: Quesnel *et al.* [32], [33] presented a real-time biosignal quality analysis of ambulatory ECG signals for detection of myocardial ischemia. The SQI based on the ensemble averaging of PQRST complexes was presented to detect unreliable ST-segments causing false alarms. The method was evaluated using the ECG signals corrupted with motion artifacts taken from physionet's long-term ST and noise stress test databases. The method had a probability of correct classification of 0.89. It is noted that the method may provide better SQA result for the ECG signals with time-varying PQRST complexes and irregular heart rates.

3) Automatic ECG Quality Scoring Method: Johannesen and Galeotti [34] presented a two-stage algorithm, which first discards the ECG recordings with macroscopic errors (missing lead, large amplitude shifts, or device saturation) and then subsequently classifies the ECG noises. The baseline wanders, powerline, and muscle artifacts were detected using the high-pass filter and cubic spline function. This method is evaluated on the 1500 annotated ECG signals taken from the PICC database. The method had an accuracy of 92.3% in the train-set and 90% in the test-set.

4) Improved Method Using QRS and T-Wave Detection: Tat *et al.* [35] designed a method to determine the quality of ECGs and its usability using an adaptation of the modified Tomkin's QRS detection algorithm followed by a set of thresholding rules based on the HR range (30–210 beats per minute), the R-peak drop rate, the standard deviation of the amplitude of the ECG data across all channels and the noise detection threshold. The method had a best SQA result of 92%.

5) ECG Quality Metrics for Clinical Trials: Vaglio *et al.* [37] presented an application specifically designed to assess quality of large number of ECG signals on the basis of analytical metrics directly computed from the digital waveforms (such as noise content, HR regularity, repolarization complexity) and parametric metrics derived from ECG measurements (such as R- and T-wave amplitudes, QT/QTc intervals).

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TABLE II
FIDUCIAL FEATURES AND HEURISTICS RULES (FF-HR) BASED SQA METHODS

Method	Signal	Event Detection	Features-Based Rules	Classes	Database	Performance
Orphanidou, 2014 [31]	ECG and PPG	R-peak	Rules based on HR, RR intervals, ratio of RR intervals and template matching (TM)	Good/Bad	PICC, JR-HA	Sp = 97%, Se = 93%
Quesnel, 2013 [32]	ECG	QRS Complex	PQRST averaging of segmented beats	Good/Bad	PLTSTD, MITBIHNST	Pcc = 0.89
Quesnel, 2014 [33]	ECG	PQRST shape	Ensemble averaging of PQRST complexes	Clean/ Contaminated	PLTSTD, MITBIHNST	–
Johannesen, 2012 [34]	ECG	QRS complex	QRS onset, offset points for detecting the macroscopic [(MS) issues missing leads (ML) and SQA]	ROC Analysis	PICC	Ac(Tr) = 92.3% Ac(Te) = 90.0%
Tat, 2011 [35]	ECG	QRS Complex	Thresholding	Good/Bad	–	A = 92 %
Xie, 2010 [36]	ECG	QRS Complex	Simple thresholding and the average peak-to-peak amplitude, locations of the QRS complexes to detect artifact such as rail contact, HF noise, low-power mask, sudden amplitude variation	Trend detection	Real time ECGs using TMC-HM	–
Vaglio, 2010 [37]	ECG	QRS Complex	HR, QRS Regularity cross-correlation criteria, median beats, amplitude and interval parameters, QRS onsets after filtering, T-wave offset position,	Good/Bad	–	–
Wang, 2002 [38]	ECG	QRS Complex	Area differences between successive QRS complexes	Histogram and CHP	MITBIHA	–
Hayn, 2012 [39]	ECG	QRS amplitude, regularity of RR intervals	Basic signal properties (amplitude, spikes, constant signal portions), number of CP in among various leads, and QRS amplitude versus noise-amplitude ratio	Good/ Bad	PICC	Ac(Tr) = 93.3% Ac(Te) = 91.6% and 0.834 (Event 2) 0.873 (Event 3)
Liu, 2011 [40]	ECG	RR interval	Four flags for detecting misplaced electrode, huge impulse, strong Gaussian noise, error of R-wave peaks by the TM after calculating Std and TH	Good/Bad	PICC	Se = 90.67% Sp = 89.78%
Hayn,2011 [41]	ECG	QRS Complex	QRS amplitude versus noise-amplitude ratio ECG amplitude, spikes, constant signal portions, number of CP in between different leads	Good/ Bad	PICC	score = 0.916 (Event 1), score = 0.834 (Event 2) and score = 0.873
Castiglioni, 2011 [42]	ECG	QRS Complex	Ratio between the power of the main cepstral peak and the total cepstral power between 0.05 and 3 s, RR interval	Classification in spectrum	Recorded using wearable MagIC device	99.8% (subject 1) and 60% (subject 2)
Johannesen, 2011 [43]	ECG	QRS onset, average RR interval	Constant derivative, amplitude TH	Excellent, Good, Adequate, Poor or Unacceptable	PICC	Se = 91% and Sp = 85%

Note: CHP: cumulative histogram plots; TMC-HM: teleMedCare health monitor; Pcc: probability of correct classification; CP:crossing points; Tr:training set; Te:test set; TH:threshold; Std: standard deviation

6) ECG Quality for Multilead Arrhythmia Analysis: Wang [38] proposed a new signal quality measure that is responsive to combinations of both physiological and nonphysiological noises. The quality measure is based on the area differences between successive QRS complexes. The mismatch metric was used for measuring the waveform area difference between two complexes. The performance of the method was evaluated on the MITBIHA database, in terms of false positive rates for premature ventricular contraction (PVC) detection. The method had an averaged premature ventricular contraction false positive rate of 0.47% for leads that were selected by using the method, and 2.56% for leads that were not selected. Although the method was shown to accurately assess the ECG signal quality, it highly demands accurate and reliable R-peak detection and QRS complex extraction approaches, which are still challenging problems under ambulatory ECG recordings.

7) QRS Detection Based ECG Quality Assessment: Hayn *et al.* [39] implemented four ECG quality measures such as empty lead criterion, spike detection criteria, lead crossing point criterion, and robustness of QRS detection for quality assessment of single lead as well as 12 lead ECG signals. The QRS detection based measure is based on the special QRS detection

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algorithm with adaptive thresholds. The quality measures were evaluated using the PICC database. The method had correct classification of 0.933 in the train-set and 0.916 in the test-set.

8) Integrated SQI Based ECG Quality on Mobile Phones: Liu *et al.* [40] proposed a method by combining several algorithms to calculate single SQI (SSQI): the algorithms to detect the straight line and huge impulse situations; the AR model analysis algorithm; the sample entropy (SampEn) analysis algorithm; and the R-peak detection and impulse rejection filter (IRF) to detect the errors in RR intervals. The integrated SQI (ISQI) was computed for 12-lead ECGs based on the 12 SSQIs. The range of ISQI is between 0 (bad quality) and 12 (good quality). The method had a sensitivity and specificity of 90.67% and 89.78%, respectively.

9) Multiple Criteria Based ECG Quality on Mobile Phones: Hayn *et al.* [41] developed a measure for mobile ECG quality assessment based on the basic signal quality properties [signal amplitude (A1), spike portion (A2), zero-line portion (A3), total length of remaining signal (A4)]; the number of lead crossing points (B); the quality of QRS detection (C); and the quality of second worst channel (D). The ECG signals were classified as "*unacceptable*" if Criterion B was fulfilled, *OR* if Criterion C was fulfilled, *OR* if any of criteria A1–A4 was fulfilled for any channel *AND* Criterion D was fulfilled. The Android algorithm achieved a score of 0.834 and score of 0.873 on the PICC database for different types of events.

10) Cepstral-Based Online ECG Quality Quantification: Castiglioni *et al.* [42] presented a cepstral-based method for online quantification of ECG quality based on the estimates of the ECG cepstrum on a running window of 10 s and the power of the highest peak in the "quefrency" band of the mean cardiac interval (between 0.25 and 2.0 s). The *quality index (QI)* is defined as ratio between the power of the cepstral peak and the total cepstral power. The QI threshold for ECG acceptability was set to 0.40, which had acceptable quality 99.8% (subject 1) and 60% (subject 2) of the tested signals as compared with an accuracy of 100% (subject 1) and 67% (subject 2), respectively.

11) Rule-Based ECG Quality Metric on Android Phone: Johannesen [43] presented an algorithm based on previously established quality metrics. The algorithm was developed based on the features such as lead-fail in all leads, global high frequency noise (filtered with tenth-order 50 Hz FIR filter), leads with noise causing QRS detection problems (average RR interval), global low frequency noise (detecting onset of QRS), and low- and high-frequency noise in the beats of sinus origin and a simple rule based approach. Evaluation results on the PICC database showed that the algorithm achieved a Se of 91% and a Sp of 85% for the training data set.

B. Fiducial Features and Machine Learning Based Methods

Table III provides a summary of fiducial features and machine learning (FF-ML) based SQA methods. These methods first compute sets of morphological and interval features from the ECG signals and then train a classifier for different types of ECG signals and noise patterns.

1) Relevance Vector Machine Based Method: Li and Clifford [44] proposed a method for false alarm (FA) reduction in the ICU monitor. The method was based on the 27 SQI metrics using 87 extracted features such as the amplitude of PPG, HR, systolic, diastolic, mean, and pulse blood pressure, difference of beats. The method achieved the maximum FA reduction rate of 30.5% with a true alarm (TA) suppression rate below 1% on the MIMIC II database.

2) Parzen Window Supervised Classifier Based Method: Redmond *et al.* [45] presented ECG signal quality measures for unsupervised telehealth environments. This SQA method consists of QRS complex detector, heartbeat waveform feature extraction, and Parzen window supervised statistical classifier. The method was evaluated on the single-lead ECG recordings. The method had an accuracy of 78.7%.

3) Machine Learning Approaches for ECG Quality: Clifford *et al.* [46] studied SQIs and data fusion for determining clinical acceptability of ECGs using the six SQIs (72 features for 12 leads) and four classifiers such as linear discriminant analysis (LDA), Naive Bayes (NB), support vector machine (SVM), and multilayer perceptron (MLP) neural network to classify the ECGs as "acceptable (1)" or "unacceptable (-1)." Results on the PICC database showed that the SVM and MLP achieved the best accuracies of 99% on the training data (Set-a) and 95% on the test data (Set-b).

4) Correlation and Neural Network Based Method: Naseri and Homaeinezhad [47] proposed a method based on three major stages: preprocessing, energy-concavity index (ECI) analysis and correlation-based examination with a suitably trained neural network. The method achieved the best Se of 100% and PPV of 98.92% for detecting high-energy noise and the Se of 92.36% and PPV of 94.77% for recognizing any other kind of disturbances using the PICC database.

5) SVM-Based Method: Behar *et al.* [48] presented a method to assess the ECG quality for both normal and abnormal rhythms for false arrhythmia alarm reduction of ICU monitors. The method employed seven SQIs introduced in previously published works and the SVM classifier with a Gaussian kernel. The SQIs are as follows: the relative power in the QRS complex (pSQI); the relative power in the baseline (basSQI); the skewness of the distribution (sSQI); the kurtosis of the distribution (kSQI); the percentage of matching of detected beats by the *eplimited* and *wqrs* algorithms (bSQI); the ratio of number of beats detected by the *eplimited* and *wqrs* algorithms (rSQI); and the ratio of the sum of the eigenvalues associated with five principal components to the sum of all eigenvalues pcaSQI [46], [48], [49]. The method was evaluated using three databases: the PICC database, the MITBIHA database, and the MIMIC II database. By using the SVM classifier with a Gaussian kernel function, the method achieved a classification accuracy of 99% for ECGs with normal sinus rhythm and accuracy up to 95% for ECGs with arrhythmia. In [49], the authors presented a single channel ECG quality metric using six SQIs and SVM classifier with a Gaussian kernel. The best results achieved on the extended database were Ac $= 0.979$, Se $= 0.976$, Sp $= 0.981$ on the training set and Ac $= 0.971$, Se $= 0.977$, Sp $= 0.965$ for the test set. It was noted that many of the records that were labeled

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TABLE III
FIDUCIAL FEATURES AND MACHINE LEARNING (FF-ML) BASED SQA METHODS

SQI Method	Signal	Event Detection	Features	Classifier	Classes	Database	Performance
Li, 2012 [44]	ECG, PPG, and ABP	QRS complex	Beat by beat onset points, amplitude of PPG, HR systolic, diastolic, mean and pulse BP, difference of beats	RVM	Excellent, Acceptable, Unacceptable	PICC, MIMIC II	FA suppression = 86.4% (A), 100% (EB) and 27.8% (ET) with no TA suppression
Redmond, 2012 [45]	ECG	QRS complex	QRS-detection-based features like MR of VEN to mean CVN, MR of VEN to median QRS height, MR of VEN to mean QRS CVN, MC of entire beat with CBS, median correlation of entire beat with clustered beat shape, MC of QRS with QRS from CBS, median correlation of QRS with QRS from CBS	PWC	Good, Average, Bad	Real ECG recordings (collected using TMC-HM)	Ac = 78.7%
Clifford, 2012 [46]	ECG	QRS complex	SED, HOS and interchannel and interalgorithm agreement, 7 SQI including the percentage of beats detected, relative power in the QRS, skewness, kurtosis, percentage of the signal to be flat line, relative power in the baseline	MLP, SVM	Acceptable, Indeterminate, Unacceptable	PICC, MITBIHNST	Ac(Tr) = 98% and Ac(Te) = 97%
Naseri, 2014 [47]	ECG	RR-interval	ECI, correlation-based examination subroutine	NN	Good/ Bad	PICC	–
Behar, 2013 [48]	ECG	QRS complex (using eplimited and wqrs)	Kurtosis, skewness, relative power in the QRS complex, relative power in the baseline, fraction and ratio of beats detected, sum of the eigenvalues associated with the five principal components over the sum of all eigenvalues obtained	SVM with a Gaussian kernel	Good/ Bad	PICC, MITBIHA, MIMIC II	Ac = 99% (Tr and Te on NSR and upto 95% for arrhythmias
Behar, 2012 [49]	ECG	QRS complex and onset points,	pSQI, kSQI, basSQI, bSQI, rSQI, pcaSQI	SVM	Classification of noise	PICC and MITBIHA, MITBIHNST	Ac(Tr) = 97.1% Ac(Te) = 93%
Kuzilek, 2011 [50]	ECG	QRS complex	TH on variance, covariance, maxima of ECG, maxima of dynamic range and average values, time lagged covariance matrices, mean, kurtosis and number of QRS complexes	SVM	Good/Poor	PICC	Score = 0.999 (Tr), 0.836 (Te)

Note: ABP: atrial blood pressure; MR: mean ratio; VEN: vector error norm; CVN: cluster vector norm; CBS: clustered beat shape; MC: mean correlation; PWC: parzen window classifier; FA: false alarm; RVM: relevance vector machine; A: asystole; EB: extreme bradycardia; ET: extreme tachycardia; Ac: accuracy; Tr: training set; Te: test set; NSR: normal sinus rhythm; GSF: Gaussian smoothing function; ASF: adaptive smoothing filtering; HOS: higher-order statistics.

good quality are classified as bad quality due to the presence of borderline or a strong baseline wander. Thus, it is difficult to distinguish between slow electrode motion and fast baseline wander. Furthermore, the method demands a broader range of baseline examples to improve the SQA accuracy.

6) Machine Learning Approaches for ECG Quality: In [50], a three stage algorithm was proposed for discarding the low quality ECG signals. In the first stage, simple features such as variance, covariance, maxima of ECG, maxima of dynamic range, and average values, were computed and compared with the threshold. Based on the thresholding result, a score was proposed. In the second stage, additional features such as time lagged covariance metrics, mean, kurtosis and number of QRS complexes were computed. These features were given to SVM for computing the score. In the final stage, the score of the first and second stages were combined to determine the signal quality. The method was tested on the PICC database and achieved a score of 0.999 and 0.836 for the training and testing sets, respectively. The authors noted that the six SQIs based on the past works are unlikely to be the optimal set of quality indicators, which may require changing for different contexts, equipment, diagnostic outcomes, or patient populations [50].

Based upon our studies and results of current state-of-the-art QRS complex detection and ECG waveform delineation methods, the practical limitations of the fiducial features based SQA methods are summarized below:

1) The QRS feature based SQA methods demand a reliable QRS complex detector under nonpathological and pathological noises that is a still challenging task under ambulatory recording environments.
2) The fiducial features based SQA methods demand accurate ECG waveform delineator for automatically determining local waveform boundaries and peaks.
3) The performance of the QRS complex shape (or ECG wave shape) based SQA methods may be degraded under time-varying PQRST morphologies. These methods may demand additional period normalization techniques under irregular heart rates.

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4) It may be difficult to find optimal decision-threshold parameters for ECG features-based heuristics rules due to large variations in pathological and nonpathological heartbeat signals and the overlapping spectra of noises and artifacts with ECG local wave spectra. In some cases, the method can provide the feature parameters within predefined decision thresholds for the noisy ECG signals.
5) Machine learning based SQA methods demand a large collection of pathological and nonpathological ECG signals and various kinds of single and combined noises in order to obtain effective signal quality group models.

C. Nonfiducial and Heuristics Rules Based SQA Methods

Table IV summarizes the nonfiducial features and heuristics rules based SQA methods. These SQA methods commonly extract sets of the time-domain, frequency-domain, time-frequency domain, statistical, and information theoretic features from the ECG signals and/or the ECG noises.

1) Multiscale Entropy (MSEnt) Based ECG Quality Assessment: Zhang *et al.* [51] analyzed the performance of the MSEnt for the assessment of ECG signal quality. The results showed that the MSEnt for LF noise and the clean ECG (quasiperiodic signal) monotonously increase with the increase of scale factor. The MSEnt for HF noise monotonously decreases with the increase of scale factor. The MSEnt for PL noise (periodic signals) exhibits fluctuation. The method was evaluated on the MITBIHA and MITBIHNST databases. The results showed that MSEnt was sensitive to the noise level within the ECG signals and could, therefore, represent a valuable tool for the assessment of ECG signal quality. This study had to first remove baseline wander from the original ECG signals because baseline wander contained in the original ECGs could lead to inaccurate results and then constructed new real ECG signals.

2) Signal Quality Estimation for Exercise ECG Recordings: Mann and Orglmeister [52] presented a method for ECG signal quality estimation under different postures and activities. The method was developed based on the PCA and the features such as the overall range between 0.2 and 15 mV, the ratio of power in frequency range 5–20 Hz to total 0–62.5 Hz, the self-correlation, correlation to averaged beat, the interlead correlation, correlation to other leads. The lowest disturbed ECG segment in the leads is automatically detected and used for reconstruction of all leads in that segment. The method was evaluated using four hours of 12-lead ECGs containing different activities including 96 min of exercise and 36 min of controlled breathing from six healthy subjects. The confidence estimation varied from 0.58 to 0.90 for a major movement to rest ECG recordings, respectively.

3) Correlation and Diversity-Based Approach: Martinez-Tabares *et al.* [53] presented an improved ECG signal quality measurement using correlation and diversity-based approaches. The unacceptable values derived from score-based criteria include the followings: the flat line (constant voltages at least for 1 s), saturation; the constant voltage superior to 2 mV by more than 200 ms; the baseline drift and channel off; the drifts over to 2.5 mV, low amplitude; and the maximum amplitude is lower than 125 μV.

4) Reconstruction Matrix Based Approach: Maan *et al.* [54] developed a ECG quality assessment method based on the correlation and Kors matrix. The method was evaluated using the PICC database and results in a correct classification of 92.2% with a Se of 97% and Sp of 75.1%.

5) Linear Prediction Based ECG Quality Classification: Noponen *et al.* [55] presented ECG quality classifier based on the residuals between filtered and observed signals, and between the best subset linear predictions without the constant term and the filtered data. The prediction coefficients are found from acceptable ECG quality using a robust method. The method achieved an accuracy of 93.2%, Se of 96.9%, Sp of 80.4%, PPV of 94.5%, and NPV of 88.3% on the PICC database. The method yielded an accuracy of 90.0% for the test data.

6) Rules-Based ECG Quality Evaluation: Moody [56] explored a number of heuristic rules that are used to detect the most common problems in ECG recordings in real time on a mobile phone. The *acceptable* signals occasionally have an extremely small or extremely large amplitude range. If the range was smaller than 0.2 mV, or larger than 15 mV, the signal was more likely to be *unacceptable*. Five rules were developed for detecting missing lead (sum over the lead is less than zero), poor contact (amplitude change of limited time duration), high amplitude artifact (high peak or absolute value of the maxima), isoline drift, and noisy lead (by threshold) in [57]. Results show that a combination of several rules is able to correctly detect a majority of poor-quality ECG signals as demonstrated using the PICC database. The method achieved a score of 0.913 and 0.896 for training and test sets, respectively.

7) Mobile ECG Quality Assessment: Langley *et al.* [58] presented an algorithm based on a set of thresholds to assess the ECG signal quality on the mobile phone. The method detects bad quality signals including flat baseline, saturation, baseline drift, low amplitude, high amplitude, and steep slope. The method was evaluated using 500 ECGs taken from the PICC database. The method had an accuracy of 85.7%.

8) ECG False Alarm Reduction Using ABP Waveform: Aboukhalil *et al.* [59] described a false alarm (FA) reduction algorithm using only one arterial blood pressure (ABP) waveform. By using the simultaneous ECG and ABP recordings, the overall FA rate was reduced from 42.7% to 17.2%.

9) Modulation Spectrum Based ECG QI: Vallejo *et al.* [60] presented a modulation spectrum based ECG QI for telehealth applications. The modulation spectral signal representation (MSSR) is used to quantify the difference between spectral variations of ECG waveforms and noise sources. The method was evaluated on 12-lead ECGs and two-lead ambulatory ECGs from the PICC and MITBIHA databases, respectively, the synthetic ECGs with varying noise levels and the real-time ECGs obtained using the Hexoskin garment during three activities (sitting, walking, running). Results showed that the method outperforms the other quality metrics such as ECG sample kurtosis and the QRS complex "in-band to out-of-band" spectral power ratio.

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TABLE IV
NONFIDUCIAL FEATURES AND HEURISTICS RULES BASED SQA METHODS

Method	Signal	Feature Based Rules	Classes	Database	Performance
Zhang, 2015 [51]	ECG	MSEnt	Good/Bad	MITBIHA and MITBIHNST	–
Mann, 2014 [52]	ECG	Overall range between 0.2 and 15 mV, ratio of power in frequency range 5–20 Hz to total 0–62.5 Hz, SF of beat by beat, ILC among leads	Good/Bad	PICC	Confidence (major movement to rest) = 0.58 to 0.90
Tabares, 2012 [53]	ECG	Autocorrelation and cross correlation among different channels and diversity due to channels	ROC	MITBIHA	Se = 86%, Sp = 91%, Ac = 90% (ROC curve)
Maan, 2011 [54]	ECG	Correlation and Kors matrix	Good/Bad	PICC	Pc = 92.2%, Se = 97%, Sp = 75.1%
Noponen, 2011 [55]	ECG	Block wise l_1-norm of the residual of linear prediction and filtered signal	Good/Bad	PICC	Ac = 93.2%, Se = 96.9%, Sp = 80.4%, Pp = 94.5%, −P = 88.3%
Moody, 2011 [56]	ECG	TH by tracking consecutive sample, overall range	Good/Bad	PICC	Score(Tr) = 0.913 Te = 0.896
Chudcek, 2011 [57]	ECG	Detecting missing lead (sum over the lead is less than zero), poor contact(amplitude change of limited time duration), high amplitude artefact(high peak or absolute value of the maxima), baseline drift, noisy lead (by threshold)	Good/Bad	PICC	0.903 for training data set A, 0.833 for testing data set B and 0.872 for evaluation data set C.
Langley, 2011 [58]	ECG	Amplitude TH to detect the flat baseline (FB); saturation (SA); baseline drift (BD); low amplitude (LA); high amplitude (HA); steep slope (SS);	Acceptable/ Unacceptable	PICC	Score = 85.7% (Te)
Aboukhalil, 2008 [59]	ECG	Some parameters such as intervals, amplitude, and gradient	Good/Bad	MIMIC II	Arrhythmia FAR (42.7% → 17.2%)
Vallejo, 2014 [60]	ECG	Difference in rate-of-change of ECG and noise sources in spectral components	Good/Bad	PICC, MITBIHA	Pearson and Spearman rank correlations.
Xia, 2011 [61]	ECG	Temporal-Spectral analysis, joint temporal-spectral analysis, self-correlation, cross correlation, and entropy analysis	Acceptable, Unacceptable and Intermediate	PICC	Ac(Tr) = 93.5% and Ac(Te) = 90.0%
Lee,2012 [62]	ECG	EMD and statistical measures, namely, Shannon entropy, mean, and variance,to calculate threshold on first IMF	ROC	Recorded using ScottCare RZ153 series	Se = 96.63% and Sp = 94.73% (both Tr and Te set), Sp = 73.66%→ 85.04% with Se (74.48%-74.62%) for AF
Oh, 2004 [63]	ECG	KLT based RMS error and residual error percent ratio of the sum of largest principal components (KL coefficients) and the trace of the eigenvalue matrix	MOS test	Europian ST-T	Se = 99.57%, Sp = 99.98%

Note: SF: self-correlation; ILC: interlead correlation; QA: quality assessment; ROC: receiver operating characteristics; Tr: training set; Te: testing set; BPF: band pass filtering; Pp: positive predictivity; Np, negative predictivity; TH: threshold; FAR: false alarm rate; KLT: Karhunen–Loève transform; RMS: root mean square; and MOS: mean opinion score

10) Computer Algorithms for Evaluating ECG Quality: Xia *et al.* [61] explored various techniques including the time domain analysis, frequency domain analysis, time-frequency analysis, self-correlation, cross correlation, and entropy analysis for evaluating ECG quality. The ECG quality was measured by the spectral radius of the matrix of regularity. Evaluation on the PICC database showed that the algorithms had an accuracy of 93.5% in the train-set and of 90% in the test-set.

11) Motion and Noise Artifact Detection in Holter ECGs: Lee *et al.* [62] presented an automatic motion and noise artifact detection in Holter ECGs using EMD and statistical features such as Shannon entropy, mean, and variance of the first intrinsic mode function (IMF). The method achieved a Se and Sp of 96.63% and 94.73%, respectively, for 15 training and 30 test data sets.

12) Karhunen–Loève transform (KLT) Based ECG Quality Measure: Oh [63] presented a method based on the root mean square and residual errors after separating the noises from an ECG signal using the KLT. The method had a Se of 99.57% and a Sp of 99.98% on the European ST-T database.

D. Nonfiducial Features and Machine Learning Methods

Table V summarizes the nonfiducial features and machine learning (NFF-ML) based SQA methods. Most methods extracted the variety of features from noise-free and noisy ECG signals. Tobon and Falk [64] presented a new ECG QI based on the MSSR. The LDA and SVM classifiers are tested to discriminate between usable and nonusable ECG segments. Morgado *et al.* [65] presented a method based on the cross-covariance matrix among various leads, the eigenvalues of the covariance

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TABLE V
NONFIDUCIAL FEATURES AND MACHINE LEARNING BASED SQA METHODS

SQI Method	Waveform	Preprocessing	Features	Classifier	Classes	Database	Performance
Tobon, 2015 [64]	ECG	–	MSSR	SVM and LDA	Good/Bad	MITBIHA, and MITBIHNST	–
Morgado, 2015 [65]	ECG	–	Cross-covariance matrix among different leads	DT and SVM	–	–	89.8%
Schumm, 2010 [66]	EDA and A-ECG	–	Crosscorrelograms, cumulative frequency plots, and the Kolmogorov-Smirnov test	SVM	Classification	MITBIHA	Se = 99.69% and Pp = 99.77%
Clifford,2011 [67]	ECG	HPF to remove BW and LF noise	Energy distribution, higher order moments and interchannel and interalgorithm agreement six SQIs: iSQI; bSQI; fSQI; sSQI; kSQI; pSQI	LDA, SVM, MLP, ANN	Acceptable, Indeterminate, Unacceptable	PICC	Ac = 99% (Set-a), 95% (Set-b) and 92.6% (test data)
Zaunseder, 2011 [68]	ECG	–	Powers in different frequency bands	Decision tree	Good/bad	PICC	Ac = 90.4%
Yu, 2006 [69]	ECG and PPG	–	HFE,LFE,LFE/HFE, PV, SNR power, entropy	SVM	Good/bad	–	92% correct classification

Note: A-ECG: ambulatory ECG; DT:decision tree; PPG:photoplethsmogram; HFE:high-frequency energy; LFE:low-frequency energy; and PV:pulse-wave variability.

matrix and the supervised binary decision tree and SVM classifier. Results on the PICC database showed that the method had an accuracy of 0.898 for the test data.

Schumm [66] presented a method for quality assessment of physiological signals during ambulatory measurements based on the crosscorrelograms, cumulative frequency plots, and the Kolmogorov–Smirnov test. Clifford *et al.* presented SQI and data fusion for determining acceptability of ECGs collected in noisy ambulatory environments. In this study, six SQIs such as iSQI, bSQI, fSQI, sSQI, kSQI, and pSQI and five classifiers (NB, SVM, MLP, and ANN) were used for assessing the signal quality. The method was evaluated using the PICC database. The method had an accuracy of 99% (Set-a), 95% (Set-b), and 92.6% (test data). Yu *et al.* [69] presented a method for automatic identification of reliable heart rates calculated from ECG and PPG waveforms. The method was evaluated using the 158 records collected during helicopter transport from trauma patients. The method achieved 92% correct classification.

E. Filtering-Based SQA Methods

Some of the SQA methods were proposed based on the filtering techniques, which were applied for assessment of noise corruption levels. A detailed summary of the filtering-based SQA methods is provided in Table VI.

1) SQI-Modified Kalman Filter (KF): Li *et al.* [71] presented an updated online system for continuously estimating HR and ABP using a SQI-modified KF. It was shown that fusing independent HR estimators from the ECG and ABP together with SQIs in a KF framework provides a significant performance improvement in tracking real episodes of extreme bradycardia and tachycardia over conventional methods used in the modern ICU.

2) Combined SQIs Based ECG Quality Estimation: Li *et al.* [72] proposed a method-based upon a novel integration of a robust KF tracking algorithm and novel SQIs for estimating HR from the ECG and ABP signals. The ECG SQA was performed by combining SQIs such as the consensus beat detection signal quality index (bSQI); the interchannel SQI (iSQI); the kurtosis-based SQI (kSQI); the spectral distribution SQI (sSQI); and the combined ECG SQI (ECGSQI).

3) ECG Signal Quality Estimators: Rio *et al.* [73] assessed eleven different methods such as: the difference between the original signal and the aligned averaged signal (average); KLT-based noise estimates; activity; mobility; complexity; turns counts; zero crossing rate; T-P interval average power divided by the QRS; cumulative mismatch histogram; first-difference histogram; and frequency content in six bandwidths and out of range event for estimating the ECG signal quality and also new three methods based on the LMS adaptive filtering, Kurtosis, and temporal dispersion. The beat detection algorithm was implemented using continuous wavelet transform based modulus maxima. It was found that Kurtosis-based quality estimator provides the best performance overall the tests. The limitation of this study is that some quality estimators need accurate QRS complex position but this is not always possible for the severely corrupted noisy ECG signal.

4) Multichannel Adaptive Filtering (MCAF) Based ECG Quality Estimation: I. Silva *et al.* [74] presented a generic sample-by-sample SQI based on MCAF that does not depend on morphological feature extraction from a target waveform. A receiver-operating-characteristics (ROC) and PR curve analysis on the simulated additive white Gaussian noise with different levels yielded overall ECG area under curve (AUC)-ROC was much lower (0.68), while AUC-ROC for the individual ECG lead ranged from 0.59 to 0.83. AUC-PR ranged from 0.23 to 0.70. Results showed that the SQI of 0.5 denotes a SNR of −15 dB whereas the SQI of 0.9 denotes a SNR of 0 dB.

5) Filters and Thresholds-Based ECG Quality Estimation: Jekova *et al.* [75] presented an ECG noise detection system to evaluate whether an episode from 12-lead ECG is reliable for diagnosis. The system used specific frequency bands to

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TABLE VI
FILTERING-BASED ECG SIGNAL QUALITY EVALUATION METHODS

SQI Method	Waveform	Features	Filtering	Classes	Database	Performance
Nemati, 2010 [70]	SQI and data fusion of ECG	Nth order spectral moment, prominent peaks, spectral purity waveform	Modified KF	Good/Poor	–	–
Li, 2009 [71]	ECG and BP	SBP, DBP, and MBP, maximum positive and negative pressure slope, maximum up-slope duration, maximum duration above threshold pulse-to-pulse interval (T), pulse pressure and ECG-ABP delay time	Modified KF	Classification	MIMIC II	Se = 99.8%, Pp = 99.3% (true BP alarms) and Se = 98.2%, Pp = 99.4% (FA)
Li, 2008 [72]	ECG and ABP waveform	Kurtosis and spectral distribution for ECG and beat by beat fuzzy logic and heuristic TH for ABP and KF for tracking after weighting each estimate	KF	Good/Bad	MIMIC II, MITBIHNST	99.9% (TTE), and 35% (FTE)
Aldecoa, 2011 [73]	ECG	Kurtosis and temporal Dispersion, correlation coefficient, beat detection, time resolution and dynamic range	LMS adaptive filtering	Good/Bad	ECG from g.Tec (g.USBamp)	high correlation 0.95 ± 0.00, Pp = 0.97 ± 0.00
Silva, 2012 [74]	ECG, ABP, PPG, Resp	Point by point SQI calculation and masking	Multichannal adaptive Filtering	Good, Bad, and May Be based on ROC	MIMIC II	–
Jekova, 2012 [75]	ECG	Adjustable TH for different noise detection (LF, HF and PLI)	Derivative and other filters with different cutoff frequency	ROC	PICC	Se = 98.7%, Sp = 80.9% (InPr), Sp = 97.8%, Se = 81.8% (TrPr)
Jekova, 2011 [76]	ECG	Amplitude dependent thresholding and rules	LPF, HPF and BPF with corresponding cutoff frequency	Acceptable/ Unacceptable	PICC	Score = 0.908 (Event 1), Se = 96.8%
Redmond, 2008 [77]	ECG	Four mask including rail contact mask, HF mask, low power mask and final combined mask to detect amplifier saturation, HF artifact, low signal power and the maximum continuous segment of usable ECG	Notch filer and HF noise by fifth-order elliptic HF filter	Good/bad	Recorded from TMC-HM	Pp = 98% & 10% (Mean & Std), −P = 97% & 19% (Mean & Std), Se = 89% & 20% (Mean & STD), Sp = 98% & 14% (Mean & Std)

Note: ABP:atrial blood pressure; TH:thresholding; InPr, incidental problems; TrPr: transient problems; KF:Kalman filter; SBP:systolic blood pressure; DBP: diastolic BP; MBP:mean BP; TTE:true tachycardiac episodes; FTE:false tachycardiac episodes; Std: standard deviation; LPF:low pass filter; HPF:high pass filter; and BPF:band pass filter.

identify the main ECG quality disruption such as missing signal or limited dynamics of the QRS components above 4 Hz; presence of high amplitude and steep artifacts above 1 Hz; BW at frequencies below 1 Hz; PLI in a band $\pm$ 2 Hz around its central frequency; high-frequency and EMG noises above 20 Hz. The method was evaluated on the PICC database. The ROC analysis showed that the method achieved high Se of 98.7% and Sp of 80.9% for any incidental problems during ECG data acquisition. Jekova *et al.* [76] presented a method for scoring the noise corruption level by evaluation of ECG amplitude dynamics, BW, PLI, EMG, and peak artifacts. For the PICC database, the method had the best accuracy of 94.2% (940/998) with higher Sp of 97.8% than Se of 81.8%.

6) ECG Quality Measures in Telecare Monitoring: Redmond *et al.* [77] presented a method based on different types of filters for automatically marking artifacts. In this study, four masks were presented for identifying amplifier saturation, HF artifact, low signal power and the maximum continuous regularity of ECG signal. The four masks are: the rail contact mask (identifies amplifier saturation by checking amplified signal in 1% of rail voltage), the HF mask (remove PLI noise by notch filter and HF noise by fifth order elliptic HF filter), the low power mask (by infinite impulse response bandpass filter of pass-band 0.7–33 Hz then square and smooth the signal by normalized 0.05 s hamming window FIR filter followed by square root operation) and the final combined signal mask (all three masks). The method was evaluated on real-time ECG signals. The method achieved a Se of 89%, Pp of 98%, and Sp of 98%. These masks do not exhibit the characteristic traits of artifact.

IV. EVALUATION OF EXISTING SQA METHODS

One of the main objectives of this review paper is to highlight the practical limitations of the SQA methods. Therefore, we implemented some of the SQA methods using the well-known Pan–Tomkins R-peak detector, which was widely used in SQA methods, the QRS detection based SQA method [39], the template matching based SQA methods [31]–[33], higher order statistics (HOS) based SQA method [81], [48] for validating the performances using the pathological and nonpathological ECG signals taken from the standard ECG databases.

A. QRS Detector Based SQA Methods

Many fiducial features based SQA methods employ traditional Pan–Tomkins QRS complex detector and its variants for automatically detection of R-peaks in the ECG signals [36], [45], [46], [49]. The Pan–Tomkins algorithm uses sets of amplitude-dependent, duration-dependent, interval-dependent thresholds

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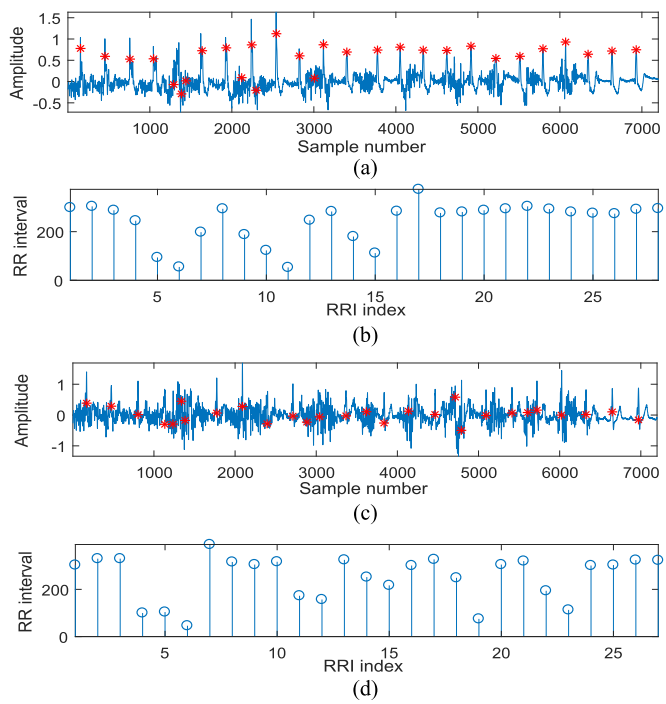

Fig. 2. Performance of Pan–Tomkins QRS complex detection for the noisy ECG signals. In some cases, the noisy RR intervals may be within the normal RR interval range that are used in the existing SQA methods.

to reject, or include the noise peaks and missed peaks, respectively. The Pan–Tomkins QRS detector can achieve better detection rates for normal ECG with regular heart rates. It is difficult to find optimal threshold parameters under time-varying PQRST morphologies and heart rates and various noises. Under time-varying heart rates and long pauses, the search-back mechanism may produce more false positives since the algorithm does not have any information about a number of true R-peaks present in a real-time ECG signal. The performance of the Pan–Tomkins QRS detector is shown in Fig. 2 for the ECG signals corrupted with muscle artifacts. This detector had more false positives under severe noises and more false negatives (FNs) for ECG signals with sudden changes in QRS complex amplitudes and shapes. The detection performance may be degraded under ambulatory and exercise ECG recordings. One can argue that if the conventional QRS detectors can detect all types of R-peaks more accurately and reliably, it can resolve the problems of some of ECG signal application systems. The results of the R-peak detection method reported in [39] are shown in Fig. 3. Although the method can accurately detect all R-peaks in the ECG signals as shown in Fig. 3(a) and (b), the method results in more FNs for an ECG signal as shown in Fig. 3(c). For different types of noise-free and noisy ECG signals, the detection performance of the method is shown in Fig. 4. Results show that the method had poor R-peak detection performance under time-varying PQRST morphologies. The limitations of the existing QRS detectors were highlighted in our previously published R-peak detection methods [78]–[80]. From our studies, it is noted that the R-peak detection errors can lead to inaccurate measurements of PQRST morphological and RR interval features, which can result in an incorrect classification of ECG quality.

B. Template Matching Based SQA Method

Some of SQA methods assess the signal quality using the similarity metric between the extracted beats and the template beat [31]–[33]. The R-peaks are first identified by using the conventional Pan–Tomkins algorithm. Then, the ECG beats are extracted by using the locations of detected R-peaks. The SNR calibration [32], [33] and correlation [31] were performed between the individual beats and averaged PQRST beat. Fig. 5 illustrates the extracted beats and the estimated ensemble beat for 10 s ECG segment as reported in [31]. From the results, it is noted that the ensemble heartbeat and all the extracted heartbeats exhibit a high correlation for the ECG signal with regular heartbeat patterns. The correlation between the ensemble heartbeat and other heartbeats is poor in case of ECG signals with different types of PQRST complexes. Therefore, these SQA methods [31]–[33] can provide good SQA performance for similar types of heartbeats but results in poor SQA performance for the dissimilar heartbeats within a quality decision making interval.

C. HOS-Based SQA Method

Some of the methods used kurtosis and variance features to detect the presence of Gaussian and non-Gaussian noises for assessing the signal quality [48], [81]. We implemented the ICA-based method, which consists of two stages: the ICA of three-lead ECGs and the kurtosis computation for all leads [81]. Fig. 6 demonstrates failure cases of ICA and kurtosis-based method in identifying the noises. In [48], kurtosis and skewness features were used for discrimination of good quality and bad quality signals. Effectiveness of these features is demonstrated for the noise-free and noisy ECG signals with muscle artifacts, power-line and additive white Gaussian noises in Fig. 7(a)–(c). Results show that the kurtosis and skewness values are higher for the good quality ECG signals as compared to the bad quality. It is noted that the kurtosis and skewness features may result in better classification accuracy for the ECG signals corrupted with additive white Gaussian noise (AWGN) and muscle artifacts and had poor result for PLI noises. However, the performance of the high-order statistics based SQA methods needs to be validated using different kinds of pathological ECG signals with various kinds of noise levels.

V. Challenging Issues and Future Directions

Based upon our past and current studies on ECG signal processing (such as R-peak detection [78]–[80], ECG compression [82], ECG denoising [16], [12], and the ECG noise detection and classification [6], [7], [83]–[86]), we highlight key challenging practical issues of existing SQA methods. We further present future research directions for the development of effective and efficient ECG signal analysis systems by significantly reducing false alarm rates under resting, ambulatory and exercise ECG recording conditions.

This review paper summarized various SQA methods for classifying the recorded ECG signals into acceptable or unacceptable. The review study shows that most SQA methods are different with respect to the feature extraction and classifi-

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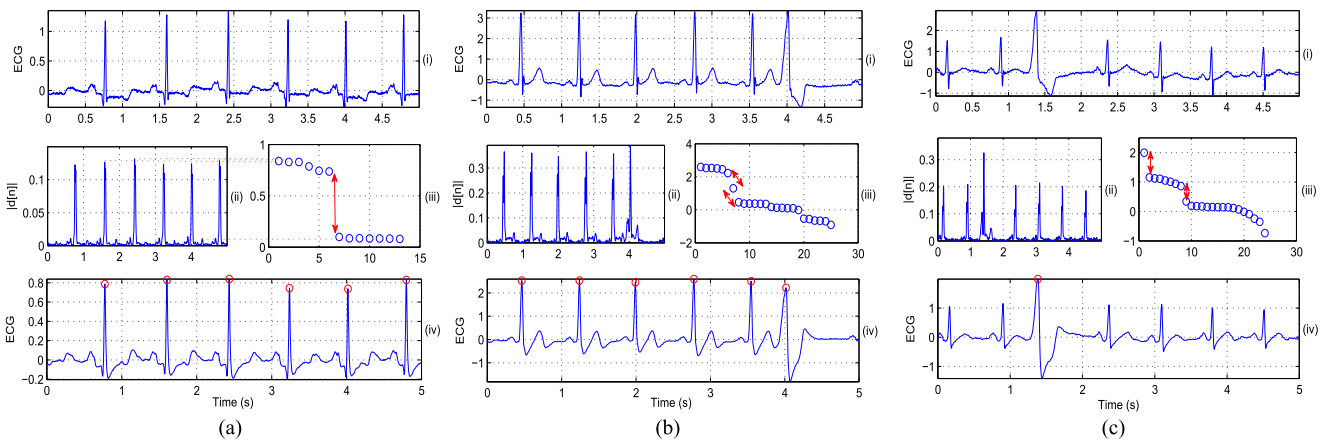

Fig. 3. Illustrates the threshold-based R-peak detection stages [39]: (i) Original ECG signal. (ii) Forward difference. (iii) Mapping of maximas using blanking function. (iv) R-peak detection results for the ECG signals as shown in (a)–(c) with different types of heartbeat patterns.

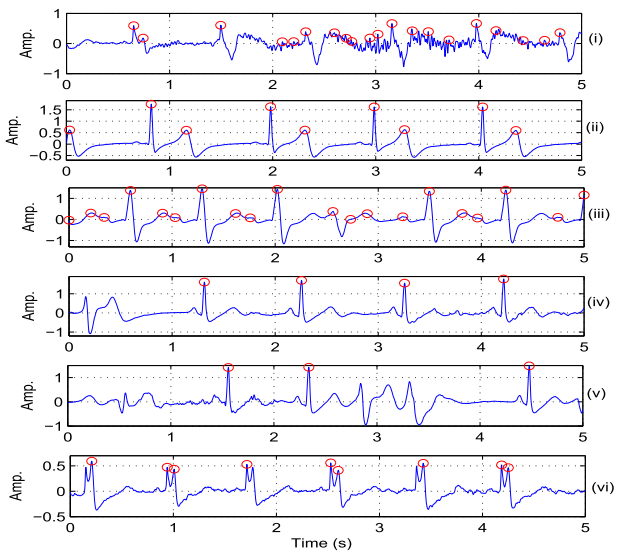

Fig. 4. Performance of the R-peak detector [39] for the ECG signals with different PQRST complexes and artifact.

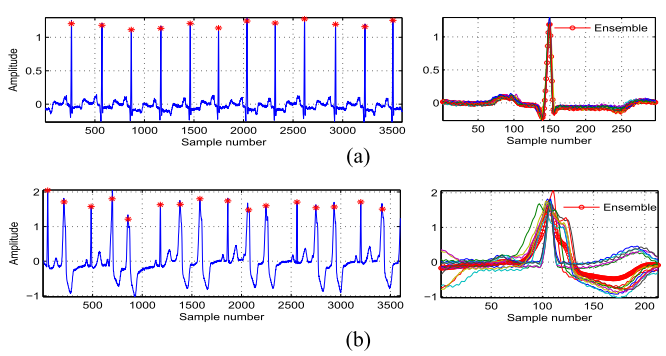

Fig. 5. Peformance of the correlation based SQA method [31]: (a) ECG signal (MITBIHA record 100). and (b) ECG signal (MITBIHA record 208).

cation technqiues. Existing SQA methods extract sets of signal quality features such as the temporal features (turning points, dynamic ranges of ECG, and prediction error) [54], [55], spectral content features (low-frequency (LF, 0.05–0.25 Hz), lower ECG bandwidth (ECG1, 0.25–10 Hz), higher ECG bandwidth (ECG2, 10–20 Hz), medium frequency (MF, 20–48 Hz), powerline noise (50 Hz, 48–52 Hz), and high frequency (HF, 52–100 Hz) [81], the time-domain and frequency-domain statistical features [46], [48], the time-frequency analysis features, local waveform morphology features (QRS-amplitude, QRS-slope, QRS-shape, QRS, and non-QRS amplitude ratio, and relative QRS spectral power), ECG segment interval features, RR interval features ($\mathrm{RR_{mean}}$, $\mathrm{RR_{std}}$, $\mathrm{RR_{max}}$, $\mathrm{RR_{min}}$, $\mathrm{RR_{max}}$ to $\mathrm{RR_{min}}$ ratio and numbers of heartbeats detected using two different R-peak detectors) [31], [39], [48], linear prediction features, eigenvalues associated with the principal components, and beat template correlation metrics. Then, the extracted signal quality features are combined at the classification stage for reducing the false alarm rates. Some studies show that different machine learning approaches are used to learn the signal and noise relationships with different noise sensitivities. These machine learning based SQA methods highly demand a large collection of datasets to adequately capture distinctive features of different morphological ECG patterns and various kinds of noises and artifacts [48]. Moreover, the feature models of certain local waves and complexes may be similar to that of the noises and artifacts. The classification accuracy of the RR interval features based SQA methods highly relies on accurate and reliable determination of heart rates using two open source QRS detectors (*eplimited* and *wqrs*). Some of the SQA methods classify the ECG signals as bad quality if heart rate (HR) is not within the range of 30–210 beats per minute [13], [31], [48], [39]. Evaluation results show that the existing R-peak detection methods had high false negative rates even for noise-free ECG signals including wide-QRS and low-QRS complexes, sudden changes in QRS morphology, tall peaked T-waves and irregular RR intervals [78].

Literature studies showed that a single signal processing technique is not adequate to remove different kinds of noises and artifacts in the ECG signals [87]. Moreover, the denoising results show that each of the filtering techniques may introduce different kinds of waveform distortion. Thus, it is very important to identify the nature of noises in the ECG recordings and

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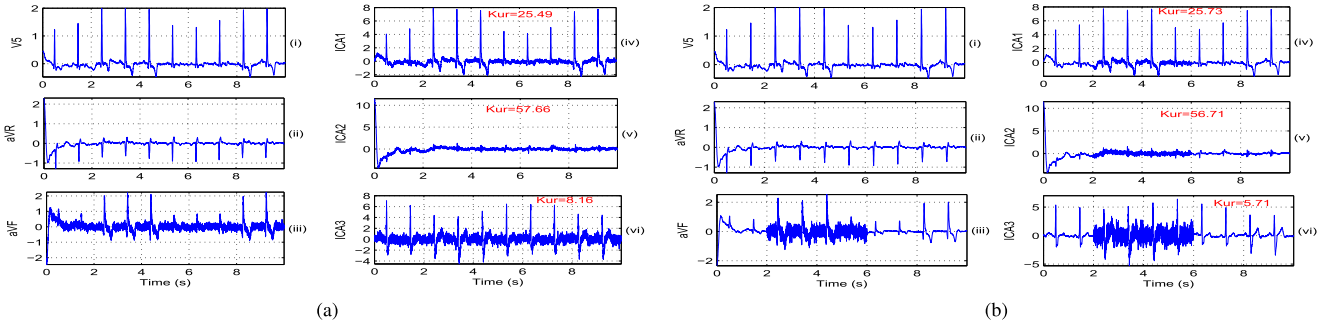

Fig. 6. Illustrates the kurtosis feature for three-lead ECG separation using the ICA [81]. (a) Noisy ECG signals. (b) Noisy ECG signals with localized noise portions in the third lead.

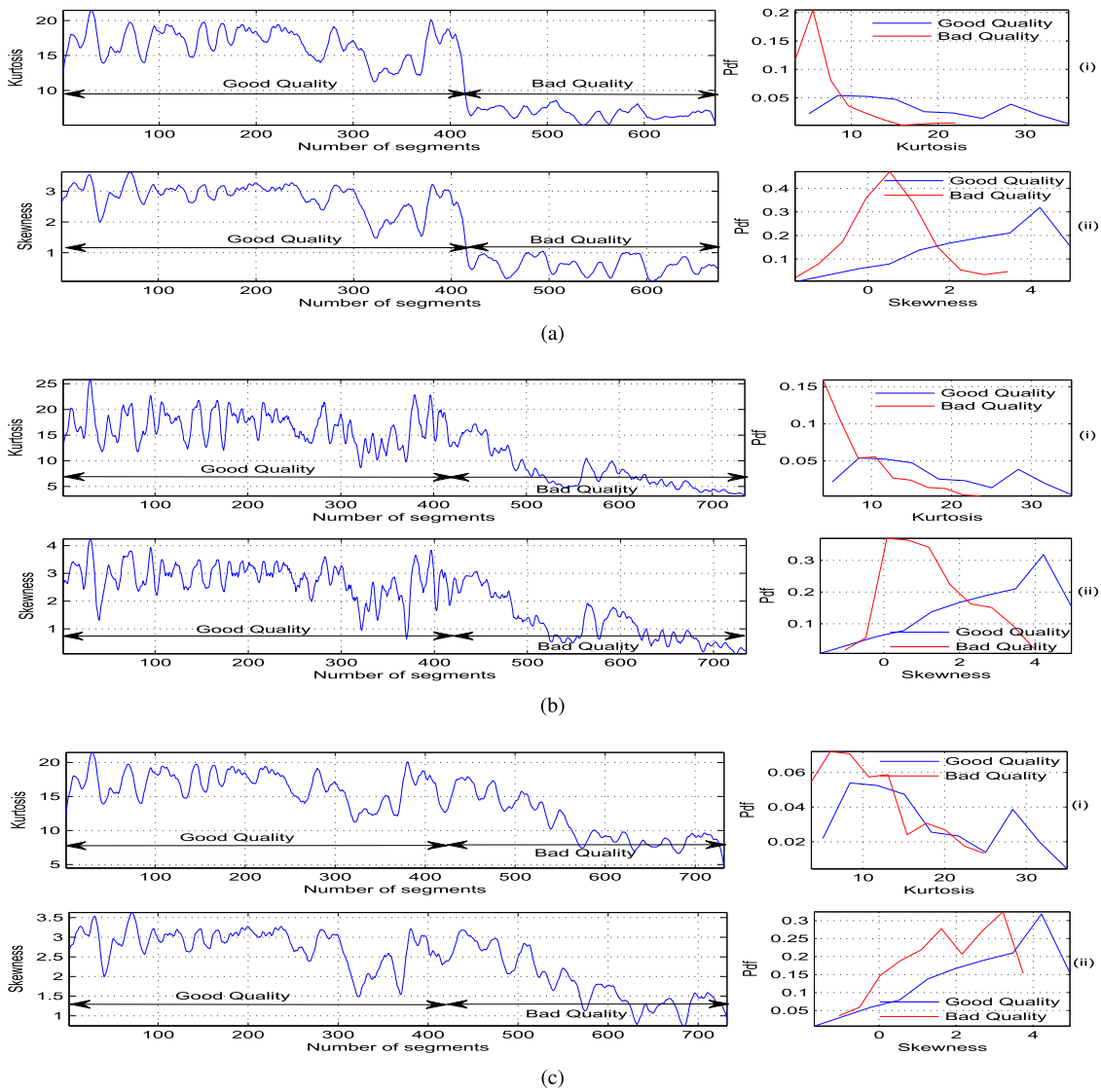

Fig. 7. Illustrates the kurtosis and skewness features and their probability density function variation for the noise-free and noisy ECG signal. (a) (i) Kurtosis feature for 413 noise-free ECG segments and 260 ECG segments with MA (ii) Skewness feature for 413 noise-free ECG segments and 260 ECG segments with MA. (b) (i) Kurtosis feature for 413 noise-free ECG segments and 322 ECG segments with AWGN (ii) Skewness feature for 413 noise-free ECG segments and 322 ECG segments with AWGN. (c) (i) Kurtosis feature for 413 noise-free ECG segments and 260 ECG segments with PLI. (ii) Skewness feature for 413 noise-free ECG segments and 260 ECG segments with PLI noise.

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to choose appropriate signal processing techniques suited to the noise types. Furthermore, in some ECG recording scenarios, the noises and artifacts can occur partially in the 10 s duration of ECG signal [7]. Therefore, it is important to determine boundaries of the localized noisy ECG portions either to mark as unreliable measurements or to discard the noisy ECG segments from the feature extraction. These aforementioned event detection issues are very important design consideration for real-time ECG signal analysis and diagnosis systems under resting, ambulatory and exercise ECG recording conditions.

Previous studies demonstrated that the contaminated ECG signals could cause classification accuracy reduction due to the R-peak detection errors (including, missed R-peaks, erroneously detected R-peaks) in the presence of artifacts or noise in the ECG signal [4]. In such scenarios, the wrongly detected beats can be classified as normal/abnormal beats incorrectly. Thus, the preprocessing stage of most methods utilizes a denoising approach to remove artifacts and noise from the ECG signal. Furthermore, each of the denoising techniques distorts different local waves such as P-wave, QRS-complex, and T-wave, and also introduces different kinds of waveform distortions such as amplitude reduction, width widening and shape alteration. In such scenarios, the normal heartbeats are misclassified as abnormal heartbeats (or life-threatening heartbeats) and vice-versa due to the R-peak detection errors (missed heartbeats, erroneously detected heartbeats) and ECG waveform delineation errors. Thus, some of the heartbeat classification methods utilized the ECG SQA strategy for distinguishing between good- and bad-quality ECG signals. However, most of the SQA methods were presented based on the ECG morphological and interval features and machine learning approaches [31]–[50]. Existing SQA methods highly demand large collections of normal and abnormal good and bad quality ECG signals to adequately capture discriminating features of different kinds of the heartbeat waveforms and the noises and artifacts.

Recent studies showed that traditional R-peak detectors had large numbers of false positives (FPs) and FNs due to the severe muscle artifacts and the sudden variations in the QRS complexes and irregular heart rates, respectively. Therefore, in most heartbeat classification methods, the presence of severe noisy ECG segments are labeled and excluded from the performance study databases. The heartbeat classification methods utilized the manually annotated R-peaks provided in the study database for extracting the ECG waveform morphological and RR interval features. This makes automated heartbeat classification methods unreliable to be widely employed for real-time unsupervised ECG diagnosis in the presence of artifacts and noises. Thus, there is still a need for the improvement of the classification accuracy, sensitivity, and specificity of the heartbeat classification method under different kinds of single and combined ECG noise sources, which can be unavoidable in ambulatory and exercise ECG recording environments.

Based upon past and present studies, we highlighted the importance of automated ECG noise analysis framework for performing real-time detection, localization, and classification of single and combined ECG noise sources. We believe that this framework has great potential applications such as: discarding severely corrupted ECG portions before ECG data transmission and processing; choosing a noise-specific signal processing scheme; and selecting optimal coding parameters for achieving efficient data compression. From the recent growths in wearable healthcare devices, it has been seen that lightweight automated ECG noise analysis scheme is highly demanded for significantly reducing the false alarms resulted from noisy measurements and boosting the battery life of wearable ECG devices that are energy constrained.

VI. Conclusion

This paper presents a review of ECG SQA methods for determining clinical acceptability of single and multilead ECG signals recorded under resting, ambulatory and exercise conditions. Many methods mainly focused on reduction of false alarms by fusing the estimated SQI values with ECG feature parameters. Some of the methods were focused on determining the unacceptable ECG signal before feature extraction. Moreover, performance evaluations were only carried out for accuracy and reliability, whereas the energy consumption aspect has often been neglected. Thus, research on real-time ECG noise detection and classifications algorithms is still in its infancy by considering the resource constrained wearable ECG monitoring devices where energy consumption is a key consideration.

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